PROJECT

DATA VISUALIZATION TEHCNIQUES

ANALYSIS OF CROP PRODUCTION DUE TO RAINFALL IN INDIA



Submitted by:

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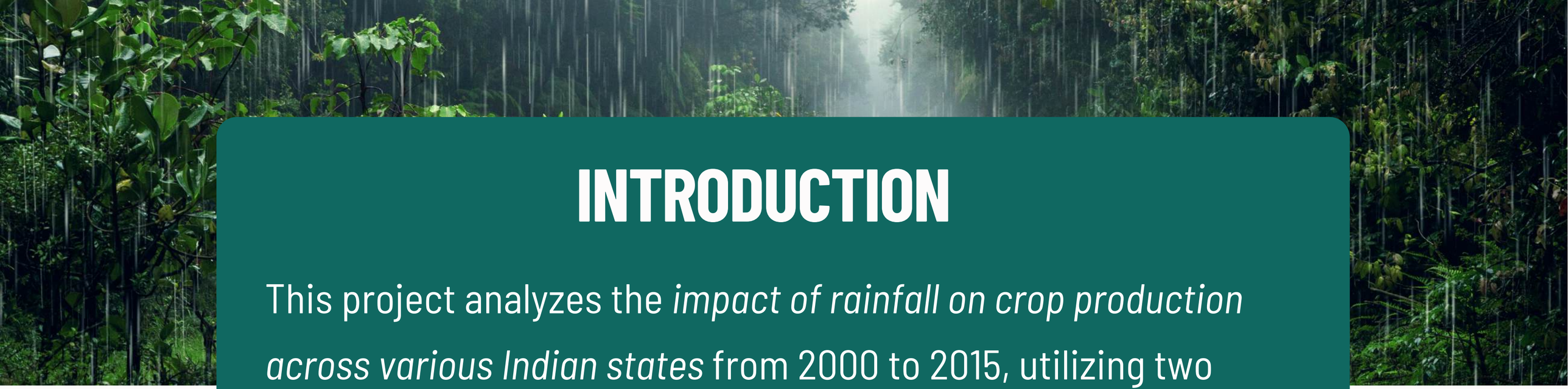
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Submitted to: Dr. Sachin Tawra Sir

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INTRODUCTION

This project analyzes the *impact of rainfall on crop production across various Indian states* from 2000 to 2015, utilizing two comprehensive datasets sourced from Kaggle. Employing Python libraries such as Pandas, Matplotlib, Plotly, and Seaborn, we aim to uncover patterns and insights into the relationship between precipitation and agricultural output.

RELEVANCE

Analyzing rainfall for its impact on crop production is crucial for several reasons.

Understanding Rainfall Patterns

- Helps identify seasonal trends and regional variability in precipitation.
- Essential for predicting droughts and flood-prone zones.
- Aids in evaluating the onset, duration, and intensity of monsoons.

Importance of Crop Production

- Backbone of the Indian economy — agriculture employs over 50% of the workforce.
- Determines food security, rural livelihoods, and GDP contribution.
- Key for understanding state-wise agricultural strengths and dependencies.

Interdependence Between Rainfall & Agriculture

- Rain-fed agriculture is practiced in over 60% of India's cultivated land.
- Crops like rice, pulses, and oilseeds are highly sensitive to rainfall variations.
- Timely and adequate rainfall ensures higher yield and reduced irrigation dependency.

Why This Project?

- To visualize how rainfall affects crop production across different Indian states.
- Identify vulnerable regions needing improved irrigation or crop planning.
- Provide data-driven insights to aid policy decisions and agricultural strategies.
- Promote awareness about the impact of climate variability on food systems.

Broader Implications

The analysis supports climate-resilient agriculture and promotes sustainable farming practices. It encourages the integration of technology and data-driven insights into agricultural decision-making. This aligns with national goals for food security and climate adaptation, ensuring a more resilient and future-ready farming ecosystem.

METHODOLOGY

We filtered and cleaned the datasets to focus on the years 2000 to 2015, enabling a clear analysis of decade-long trends and patterns in rainfall and crop production across India.

The data sets used are:



crop_yield.csv

01

This dataset offers insights into crop production across Indian states, focusing on the relationship between rainfall and agricultural yield. It includes key variables essential for analyzing regional and seasonal farming patterns.

- Crop, State, Season, Year – Define the agricultural context
- Area & Production – Land used and output produced
- Annual Rainfall – Yearly precipitation in mm
- Fertilizer & Pesticide – Input usage data
- Yield – Crop productivity (output per unit area)

The dataset is ideal for exploring how rainfall patterns impact crop yield, supporting climate-aware agricultural planning.



rainfall_in_india.csv

02

Here is a brief description of the second dataset in precise paragraph form with bullet points included:
This dataset captures monthly and seasonal rainfall data across different meteorological subdivisions in India. It is crucial for identifying regional rainfall trends and understanding long-term climate behavior.

- SUBDIVISION & YEAR – Region and time reference
- JAN to DEC – Monthly rainfall data (in mm)
- ANNUAL – Total yearly rainfall
- Jan-Feb, Mar-May, Jun-Sep, Oct-Dec – Seasonal rainfall breakdown

This dataset is key for analyzing how rainfall varies by region and season, helping assess its impact on agricultural productivity across India.



rainfall_vs_crop_production.csv

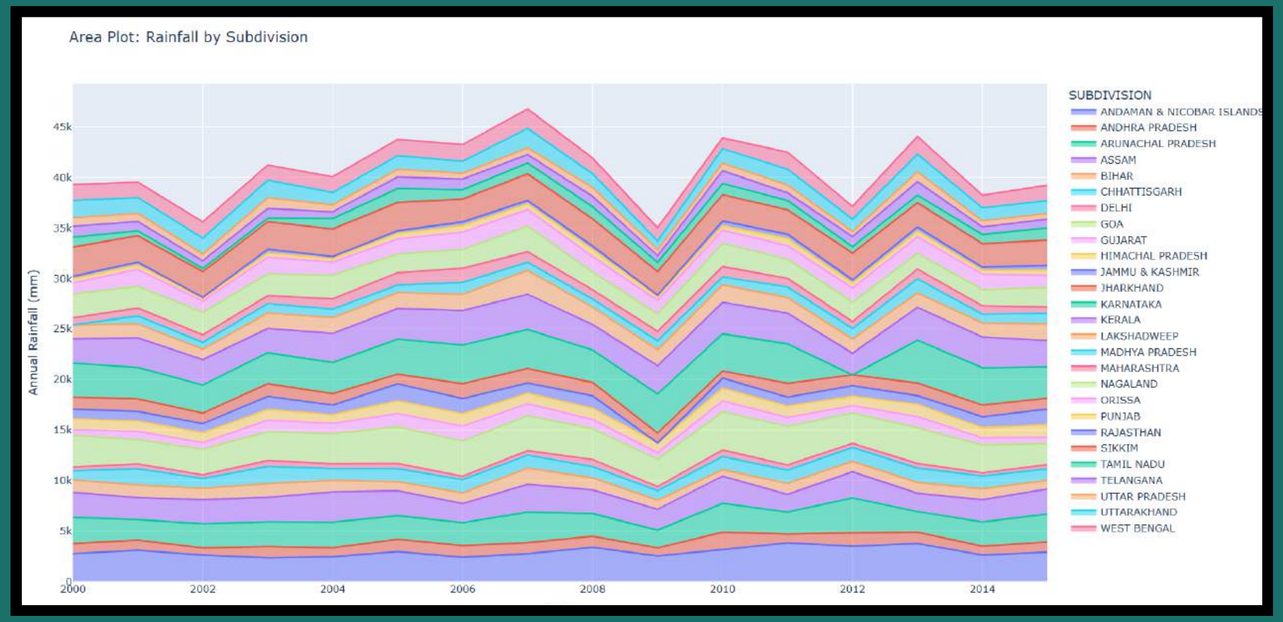
03

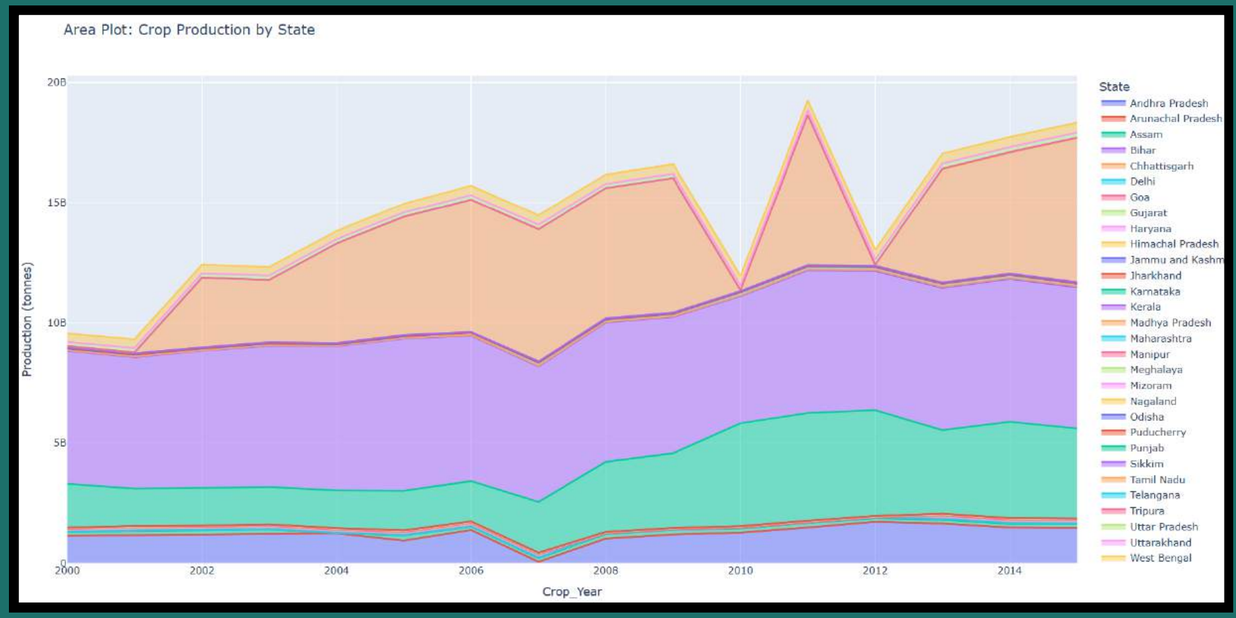
Here is a concise paragraph description of the merged dataset, along with bullet points for clarity:
This dataset is a merged version of the crop production and rainfall datasets, allowing for integrated analysis of how annual rainfall impacts crop yield across Indian states. It is tailored for visualization and correlation analysis in tools like Tableau.

- Year & State – Temporal and regional identifiers
- Crop – Type of crop cultivated
- Area (hectares) – Land area used for cultivation
- Production (tonnes) – Total output
- Yield (tonnes/hectare) – Productivity per unit area
- Annual Rainfall (mm) – Total precipitation in the corresponding year and state

VISUALIZATIONS

AREA PLOT

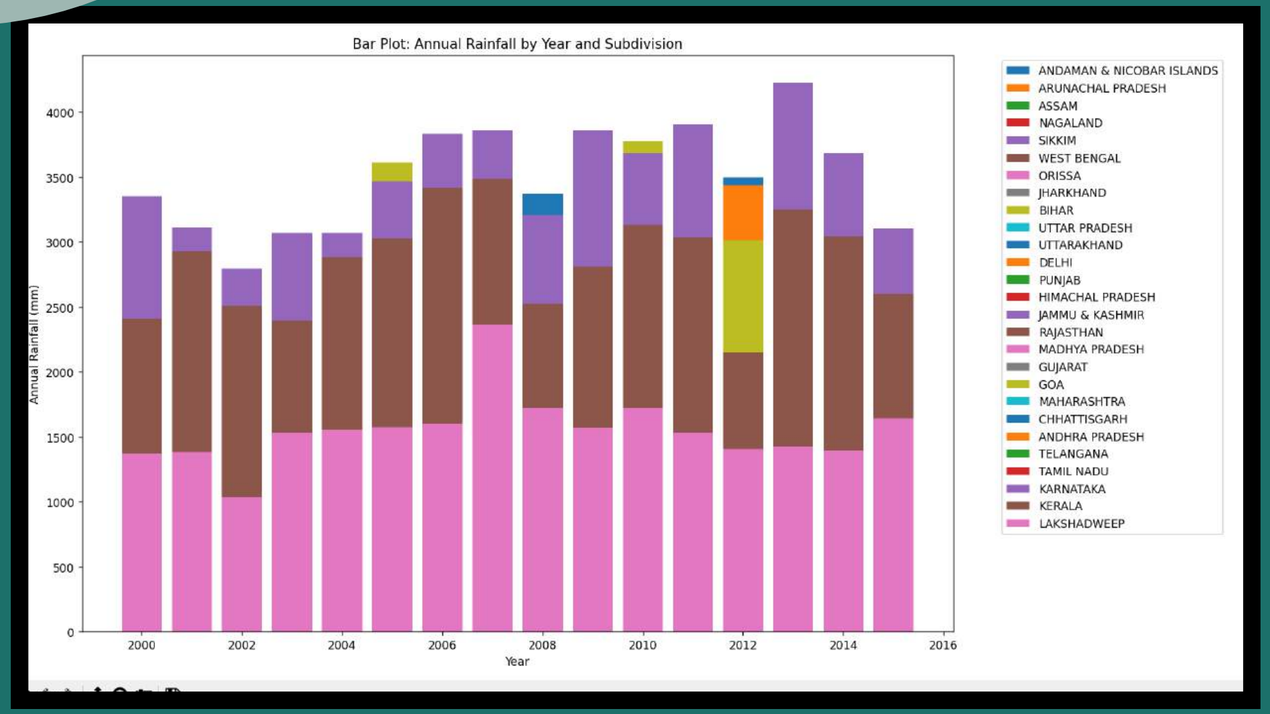


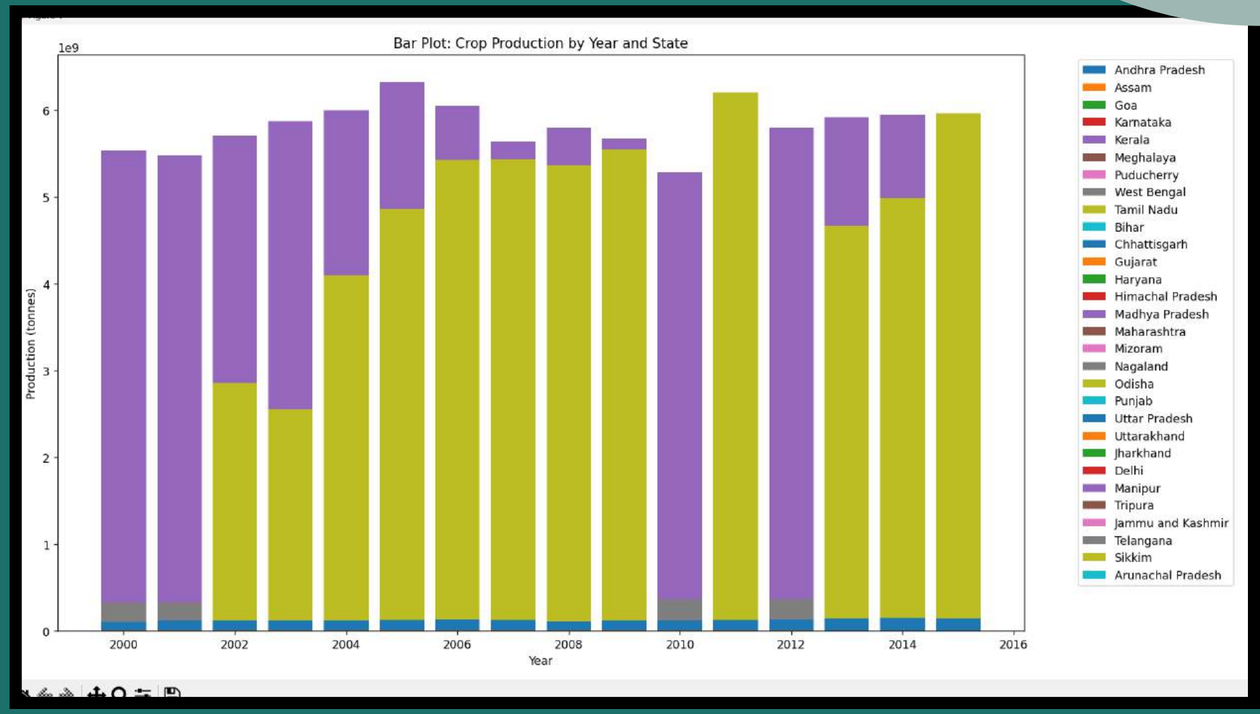


Rainfall by Subdivision

Crop Production by states

BAR PLOT

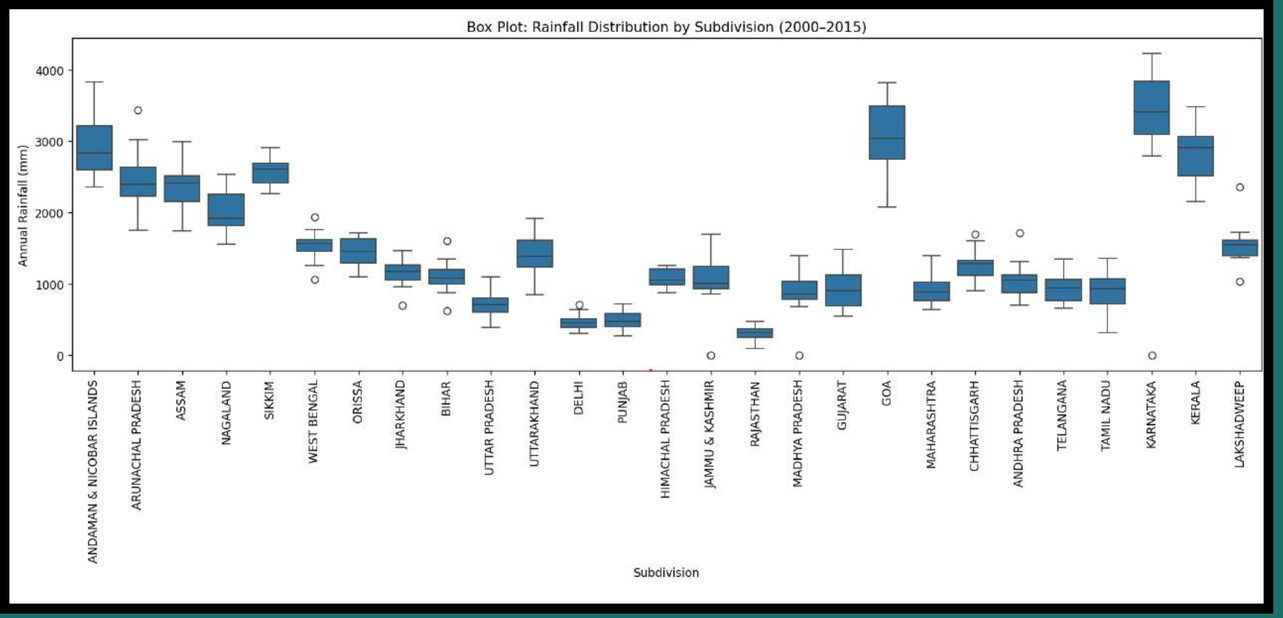


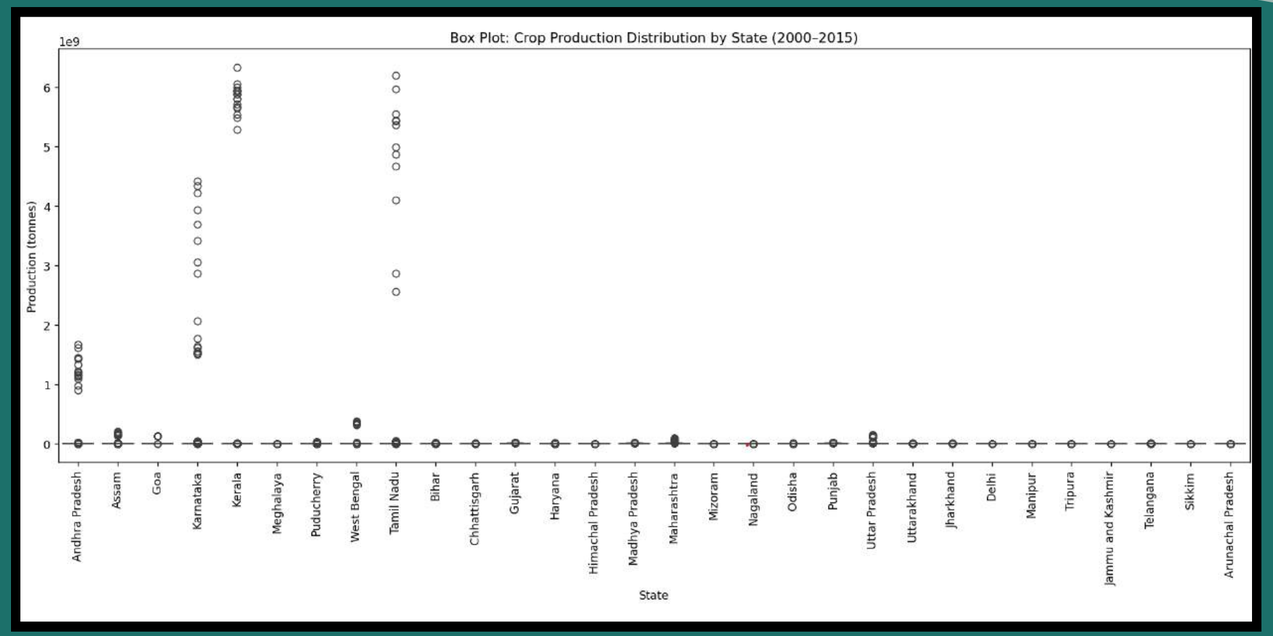


Annual Rainfall by year and subdivision

Crop Production by year and state

BOX PLOT

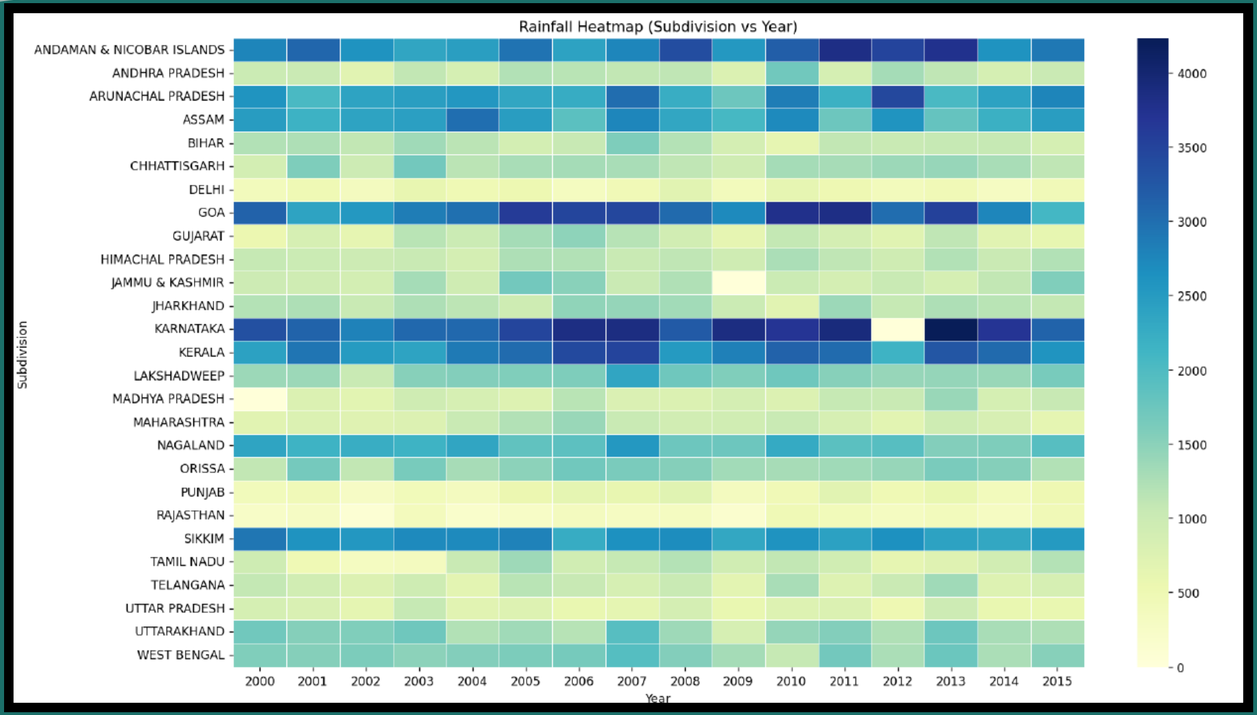


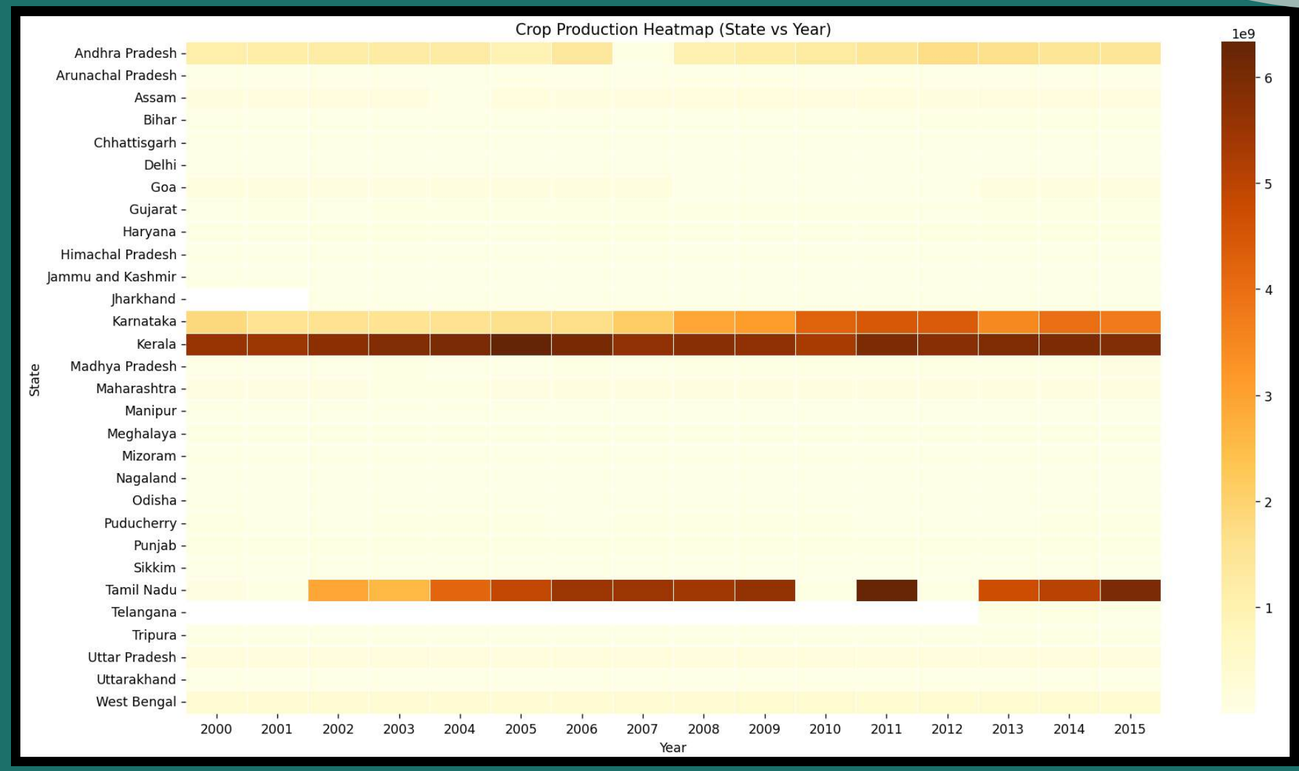


Rainfall Distribution by Subdivision(2000-2015)

Crop Production Distribution by State(2000-2015)

HEATMAPS

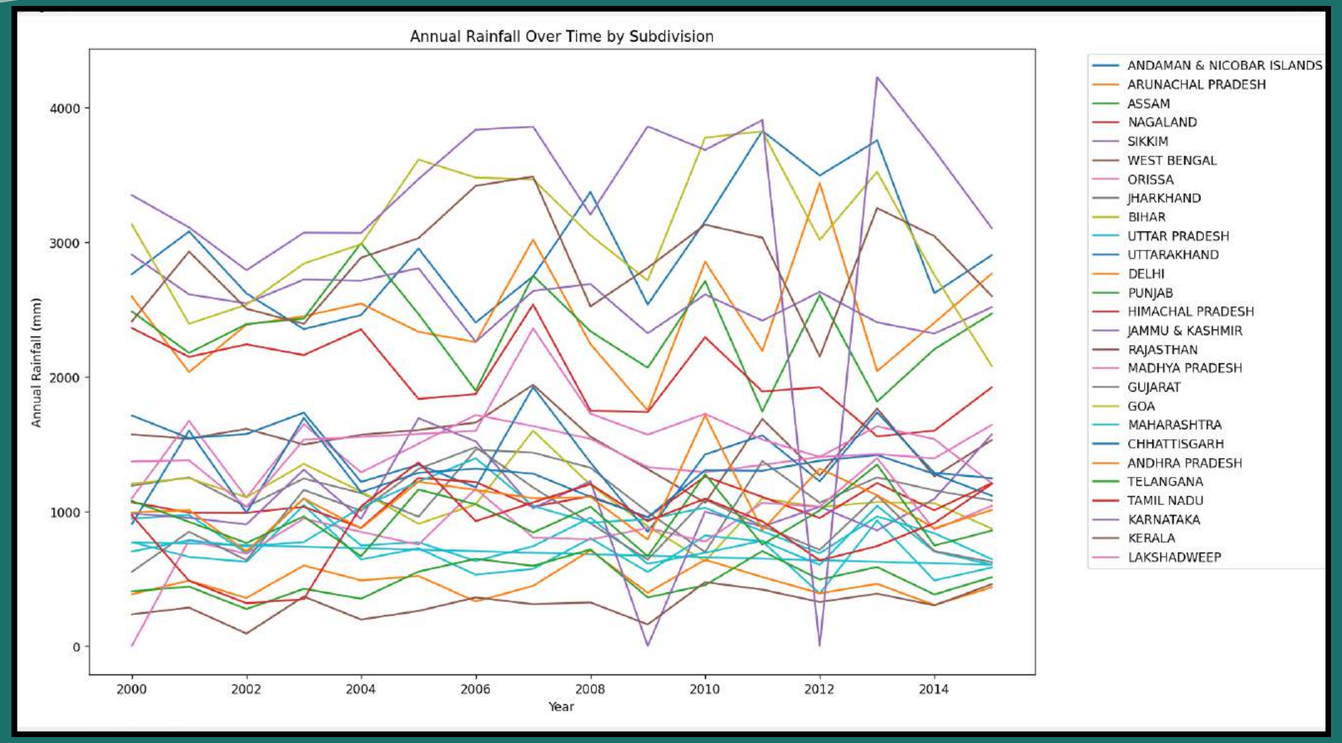


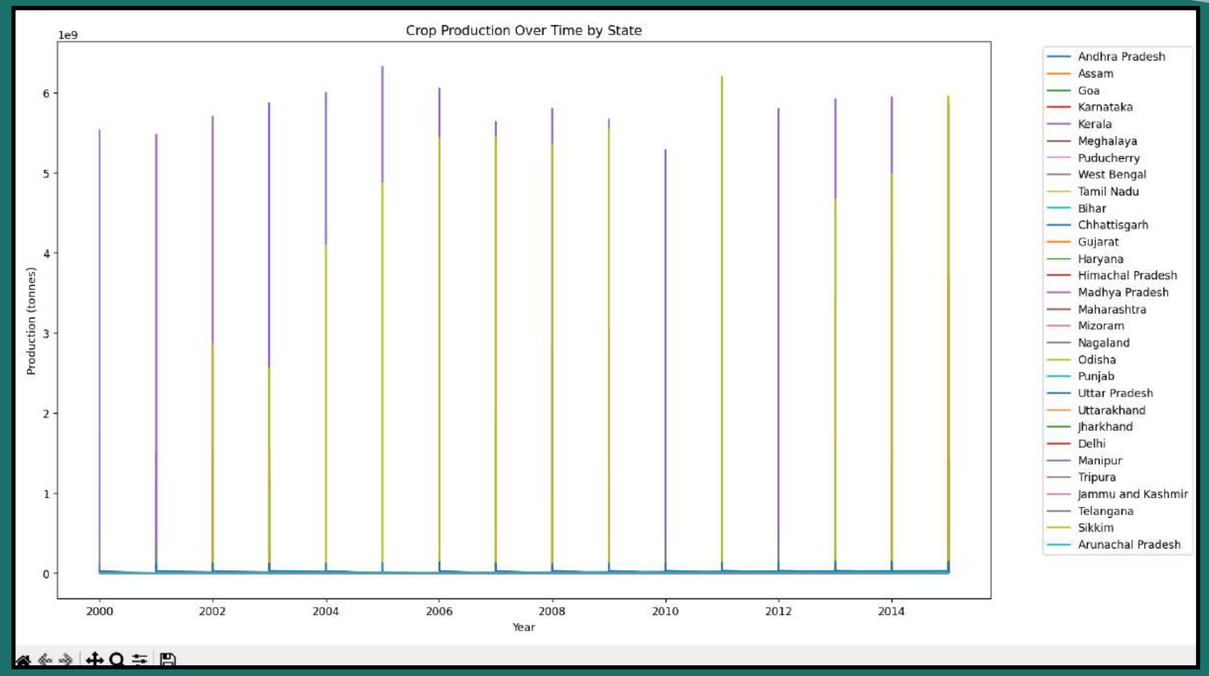


Rainfall Heatmap (Subdivision vs Year)

Crop Production Heatmap (State vs Year)

LINE PLOT



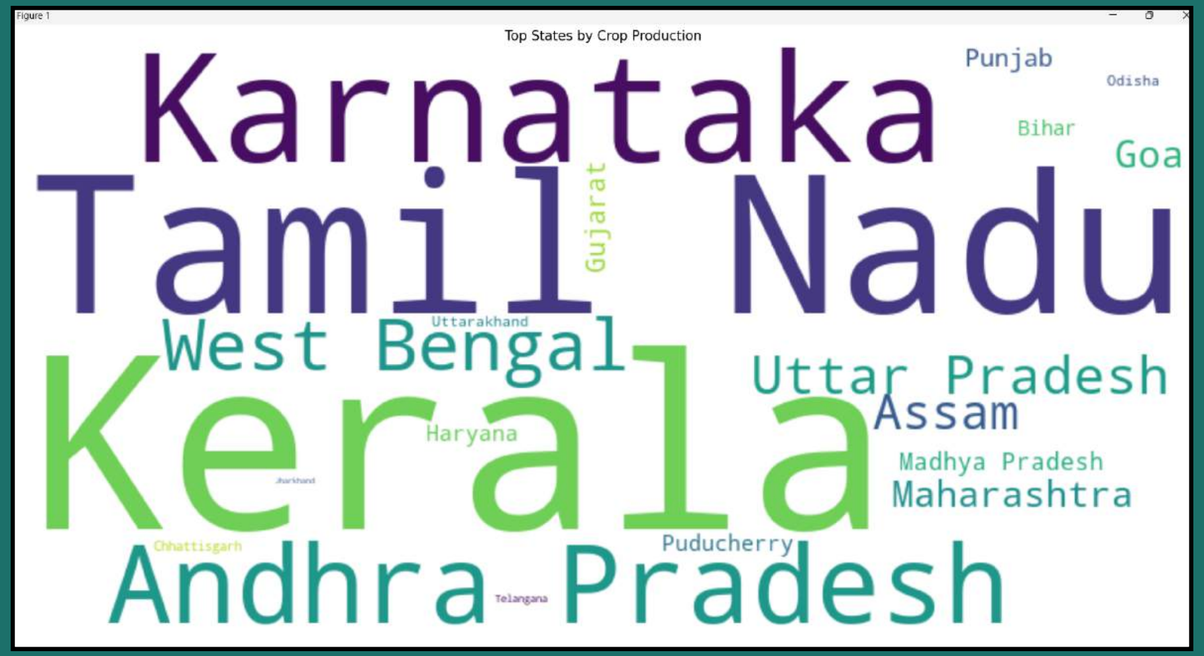


Annual Rainfall over time by Subdivision

Crop Production over time by State

WORD CLOUD

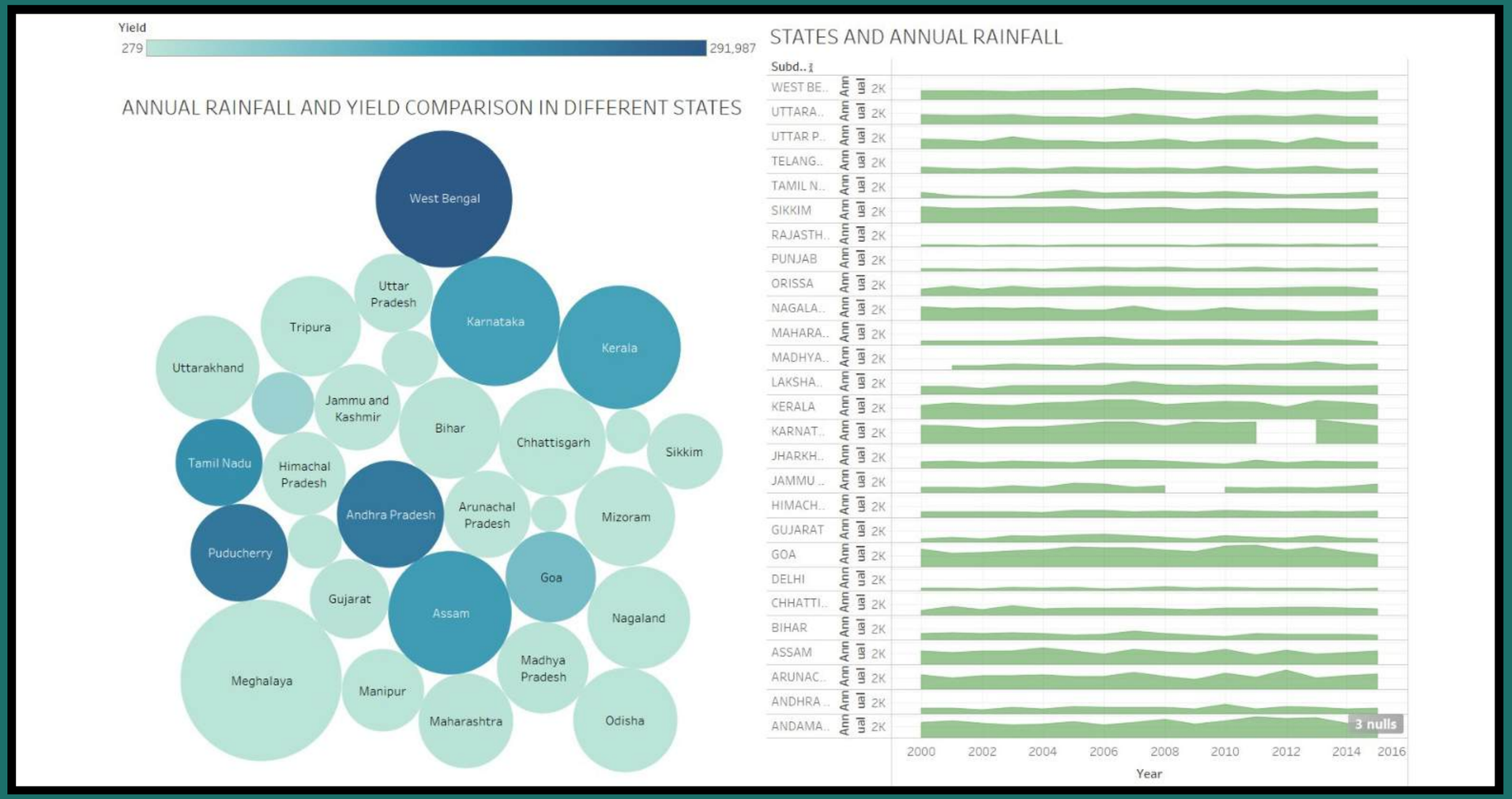




Top Subdivisions by Average Rainfall

Top States by Crop Production

TABLEAU

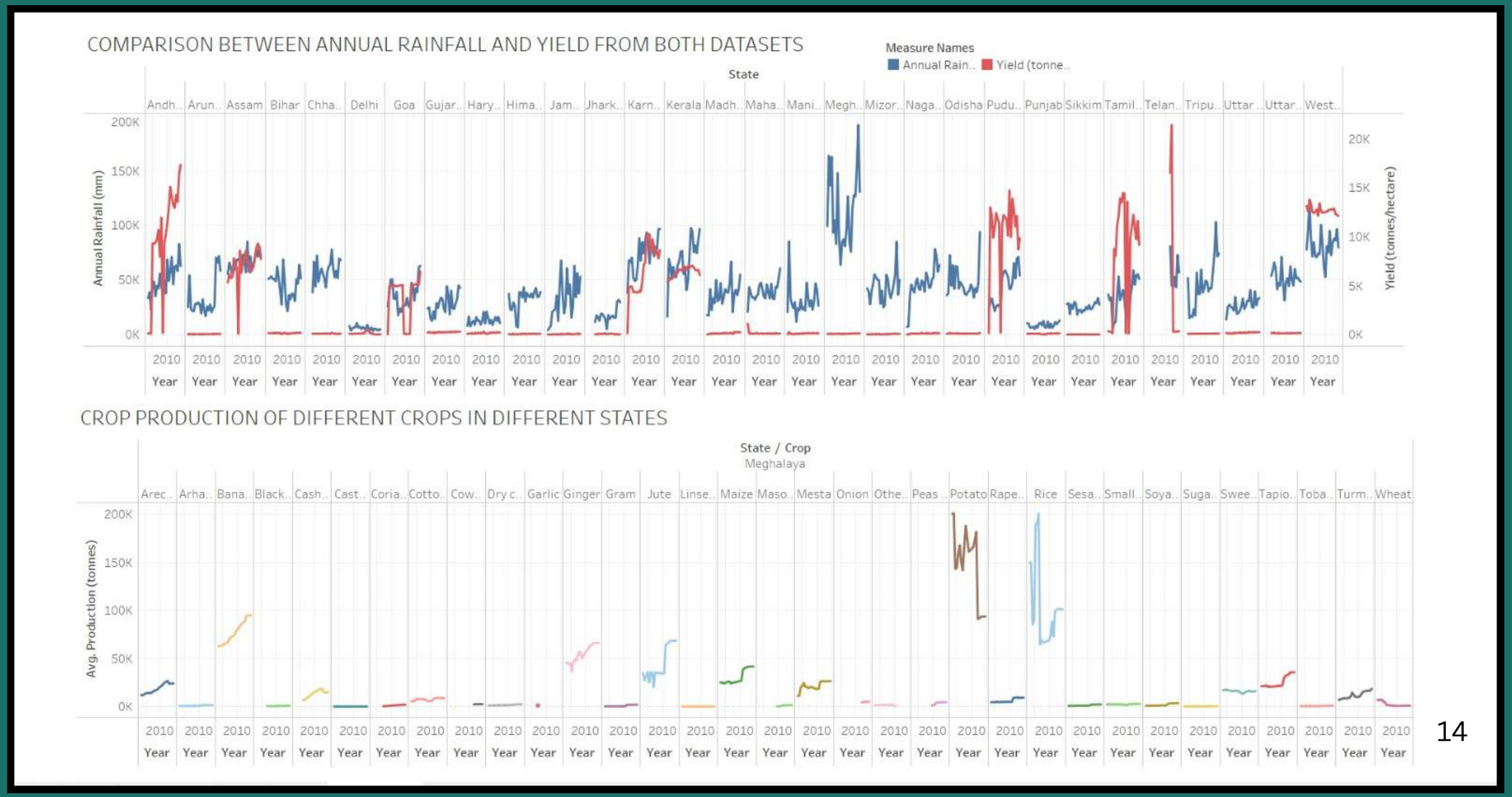


Dashboard visualising crop yield dataset(left) and rainfall dataset(right individually)

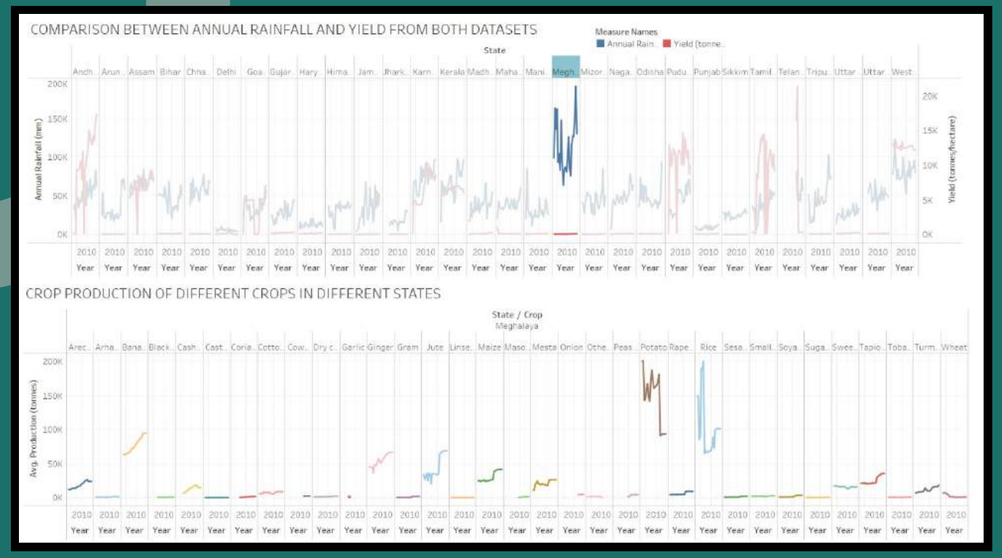
RESULTS

TABLEAU

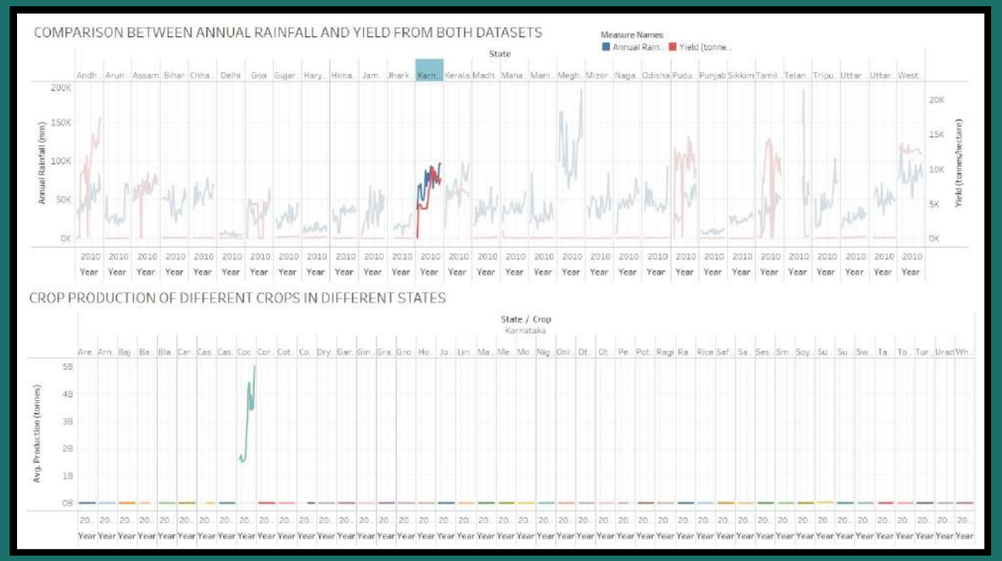
This visual dashboard presents a comparative analysis of crop yield and rainfall across different states. It simultaneously illustrates the amount of rainfall received and the corresponding crop yield produced in each state, based on data from a merged dataset.



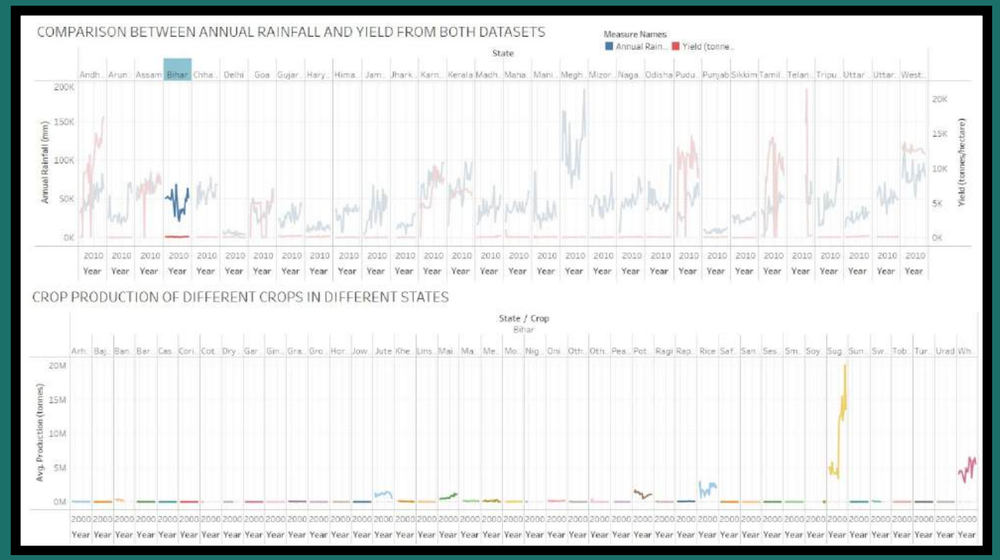
Let us now explore visual comparisons of different states as presented in the dashboard.



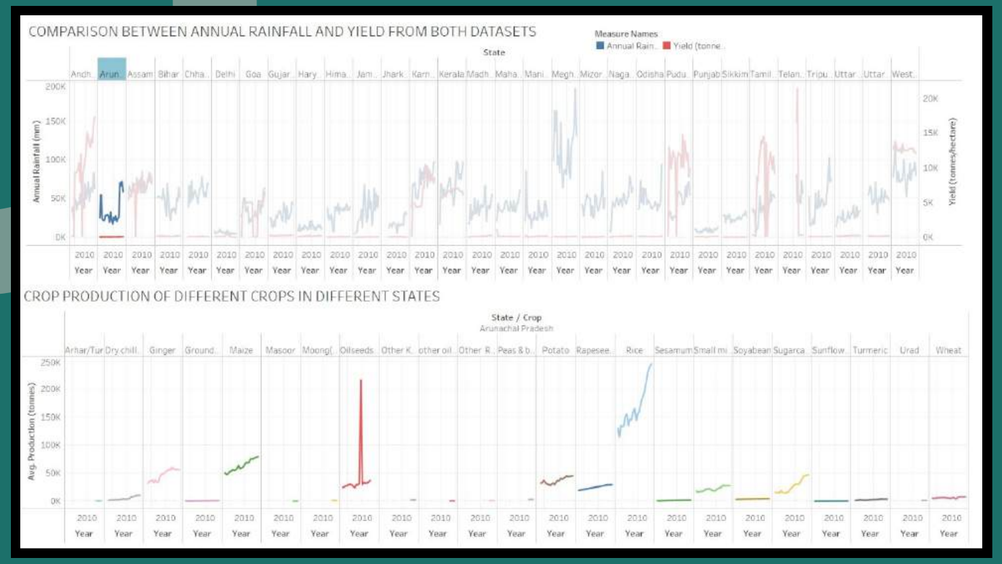
MEGHALAYA



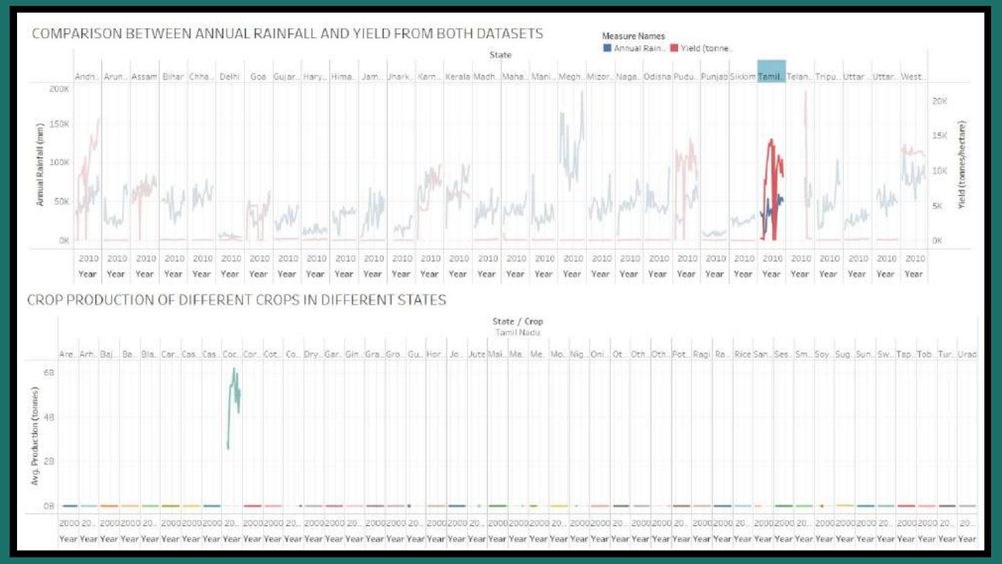
KARNATAKA



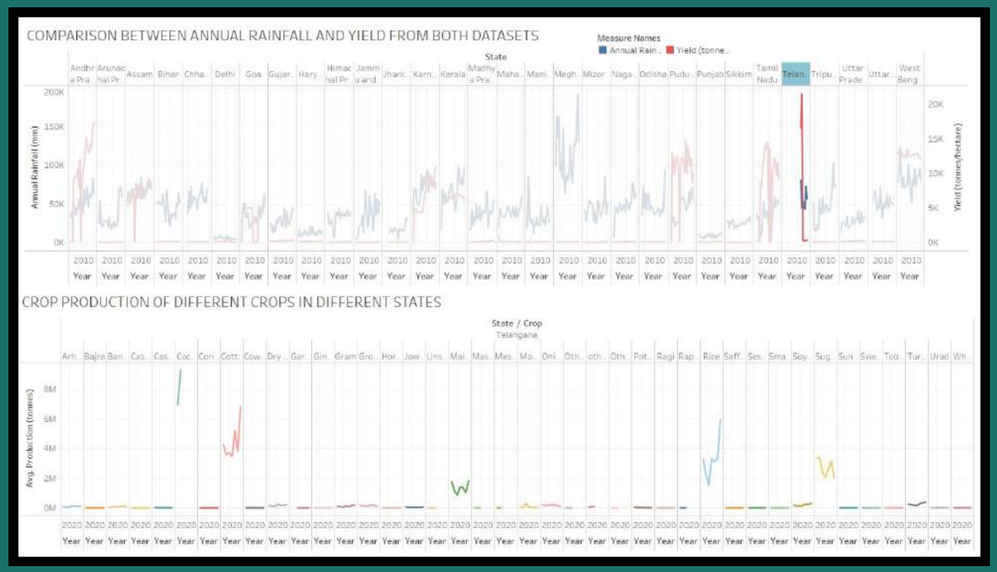
BIHAR



ARUNACHAL PRADESH

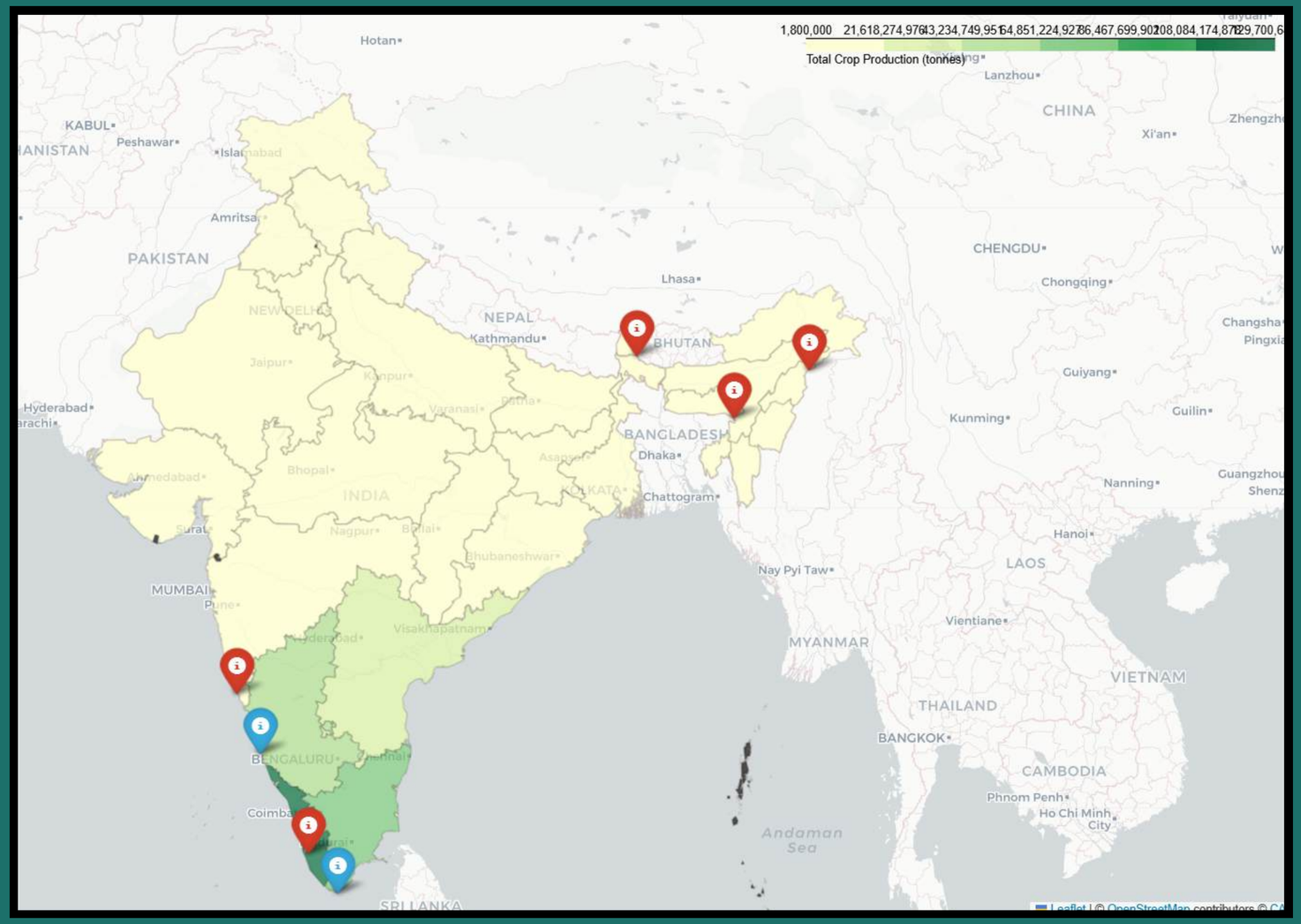


TAMILNADU



TELANGANA

FOLIUM



RED AND BLUE MARKERS (ICON PINS)

These markers indicate states of special interest, based on two key conditions:

- 🔴 Red Marker: States where the average rainfall exceeds 2500 mm. Ex. Kerala and Goa, Northeast states
- 🔵 Blue Marker: States with total crop production above the 90th percentile (top 10% of producers). Ex. Tamil Nadu, Karnataka

CHOROPLETH COLOR SHADES (YELLOW TO GREEN)

These shades represent total crop production per state, derived from the

- 🟨 Light Yellow Lower production
- 🟩 Darker Green Higher production

NOTE: Because Kerala, Tamil Nadu, and Karnataka have extremely high values (in the tens of billions), the gradient is skewed — so states like Punjab or UP may appear lighter even if they produce a lot.

CONCLUSION

How Rainfall Affects Crop Yield and Production (2000–2015)

Rainfall remains one of the most influential natural factors in Indian agriculture. Its effects on crop yield and production are significant, especially in rain-dependent regions.

01

Rainfall Drives Kharif Crop Success

- Kharif crops (e.g., rice, maize, pulses) are sown with the onset of the monsoon (June–September).
- Adequate and well-distributed rainfall during this period is crucial for germination, growth, and harvest.
- Shortfall or delay in monsoon affects sowing windows, leading to reduced acreage and lower yield.

02

Rainfall Variation Causes Yield Instability

- States like Maharashtra, Odisha, and Madhya Pradesh showed visible dips in crop production during drought years (e.g., 2002, 2004, 2009).
- These are predominantly rain-fed regions, with limited irrigation infrastructure, making them highly vulnerable.

03

Rainfall Dependency Differs by State

- High dependency in rain-fed states: Crop output rises or falls with monsoon performance.
- Moderate dependency in semi-irrigated states: Production dips slightly in drought years but generally remains stable.
- Low dependency in irrigation-intensive states: Yields remain consistent regardless of rainfall variation.

Why Rainfall Is Not the Only Factor in Crop Production

Although rainfall is critical, it is not the sole determinant of agricultural success. Several non-climatic factors significantly influence crop yield and overall production.

Irrigation Infrastructure is Crucial

States like Punjab and Haryana consistently produce high yields with moderate and stable rainfall.

Their advantage comes from:

- Extensive canal networks
- Tube well irrigation
- Water-intensive cropping patterns (e.g., rice-wheat)

Technology and Mechanization

Use of improved seed varieties, fertilizers, pesticides, and farm machinery boosts productivity even in less favorable climates.

Mechanization reduces dependence on climate, enabling timely planting and harvesting.

Policy and Institutional Support

Minimum Support Prices (MSPs), government procurement, crop insurance, and input subsidies help reduce climate-related risks.

These supports are stronger in politically prioritized agricultural states.

Crop Selection and Agro-ecological Adaptation

Inland Karnataka achieves decent yields by growing drought-tolerant crops like millets and pulses, despite low rainfall.

In contrast, North-East India, despite abundant rain, shows low yields due to unsuitable terrain and crops, and poor logistics.

SOURCES

1. CROP YIELD DATASET:
https://www.kaggle.com/datasets/akshatgupta7/crop-yield-in-indian-states-dataset

2. RAINFALL DATASET:
https://www.kaggle.com/code/darsh79/starter-rainfall-in-india-99bfc809-4/notebook

3. India Meteorological Department. (2000–2015). Annual and seasonal monsoon rainfall reports. Government of India.
https://mausam.imd.gov.in/

4. Ministry of Agriculture & Farmers Welfare. (2000–2015). State-wise crop production statistics. Government of India.
https://agricoop.nic.in/

5. *Gadgil, S., & Gadgil, S. (2004). The Indian monsoon and its variability in relation to agriculture. Current Science, 87(2), 200–206.*

6. *Government of India. (2015). Economic Survey of India 2014–15: Chapter on Agriculture and 6. Climate. Ministry of Finance.*
https://www.indiabudget.gov.in/

THANK YOU

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